Hirofumi Shiba

https://162348.github.io/
shiba.hirofumi@ism.ac.jp
Last Update: 2026/11/7

Education

- **PhD Statistical Science**. *Graduate University for Advanced Studies, SOKENDAI, Tokyo, Japan*. 2023.4 – 2028.3
  Superivsor: Prof. Kengo Kamatani
- **BA Mathematics**. *Department of Mathematics, University of Tokyo*. 2019.4 – 2023.3
  Supervisor: Prof. Nakahiro Yoshida

Publications

1. **Hirofumi Shiba**, Kengo Kamatani (2026). Diffusive Scaling Limits of Forward Event-Chain Monte Carlo: Provably Efficient Exploration with Partial Refreshment. *Submitted to the Annals of Applied Probability*.

Talks

1. 2026-10-13. (15:30-16:30) **Hirofumi Shiba**. High-Dimensional Behaviour of Two Piecewise Deterministic Markov Processes: Efficiency Comparison & Asymptotic Variance Estimation. *Bloomsbury Probability Seminar*. 1-19 Torrington Place G08, London, UK.
2. 2026-06-16. (9:00-10:40) **Hirofumi Shiba**. High-Dimensional Behaviour of Piecewise Deterministic Monte Carlo: Efficiency Comparison & Asymptotic Variance Estimation. *The Seventh Institute of Mathematical Statistics Asia Pacific Rim Meeting (IMS-APRM)*. Henry Cheng International Conference Centre, Cheng Yu Tung Building, The Chinese University of Hong Kong.
3. 2026-02-10. (9:50-9:55) **Hirofumi Shiba**. Computational Complexity Comparison via Scaling Limits for Emerging MCMC Samplers. *ISI-ISM-ISSAS Joint Conference*. Academia Sinica, Taiwan (2nd Conference Room, 3F).
4. 2026-02-05. (11:40-12:00) 司馬博文. 或る区分確定的マルコフ過程のスケーリング極限. **学生研究発表会（統数研）**. 統数研 2 階大会議室（B201）.
5. 2025-07-15. (17:30-18:00) **Hirofumi Shiba**. Understanding Discrete Denoising Diffusion Models: A Survey and a Gap in the Theory. *Workshop on Opportunity Week 2025, Stochastische Geometrie und räumliche Statistik*. Universität Ulm.
6. 2025-02-06. (16:50-17:10) 司馬博文. 動き出す次世代サンプラー: 区分確定的モンテカルロ. **学生研究発表会（統数研）**. 統数研 2 階大会議室.
7. 2024-09-10. **Hirofumi Shiba** | 司馬博文. Zig-Zag Sampler: A MCMC Game-Changer. *Talk*. Seoul National University, Gwanak (관악) campus, South Korea.
8. 2024-07-25. 司馬博文. Zig-Zag サンプラー: 物理のくびきを超える MCMC. **ベイズ会**. 本郷キャンパス小島ホール第二セミナー室.

Posters

1. 2026-10-31. (14:00-15:30) 司馬博文. Sticky Manifold PDMP による低次元多様体上に台を持つ分布からのサンプリング. **異分野異業種研究交流会**. 東京大学 駒場キャンパス I 21 KOMCEE West 地下 1 階 MM ホール.

2. 2026-07-02. (13:50-15:20) **Hirofumi Shiba**. A PDMP Scaling Analysis: Gains from Persistent Momentum. *ISBA Satellite Meeting 2026*. Institute of Statistical Mathematics.
3. 2026-06-29. (17:15-19:30) **Hirofumi Shiba**. A Provable Comparison Scheme via Scaling Limits for Emerging Monte Carlo Samplers. *ISBA World Meeting 2026*. WINC AICHI (Aichi Industry & Labor Center).
4. 2026-05-22. (13:00-15:00) 司馬博文. PDMP の拡散極限と早期収束診断. **統計数理研究所オープンハウス**. 1 階交流アトリウム.
5. 2026-03-03. (16:00-18:00) **Hirofumi Shiba**. Diffusive Scaling Limits & Early Diagnostics for PDMP Samplers. *Workshop on Functional Inference and Machine Intelligence*. TKP Shinjuku West Exit Conference Center, Shinjuku, Tokyo, Japan.
6. 2026-02-16. (15:45-16:45) 司馬博文. 区分確定的モンテカルロ法の拡散極限と早期収束診断. **第２回：計算技術による学際的統計解析ワークショップ**. Institute of Statistical Mathematics (Room D305).
7. 2026-02-10. (8:50–11:00, 11:20-12:20 (Group B)) **Hirofumi Shiba**. Computational Complexity Comparison via Scaling Limits for Emerging MCMC Samplers. *ISI-ISM-ISSAS Joint Conference*. Academia Sinica, Taiwan (2nd Conference Room, 3F).
8. 2025-11-12. (17:10-19:10 (1b-R-02)) 司馬博文. メトロポリスを超えた枠組みで我々はどこまで行けるか？: Forward Event-Chain Monte Carlo のスケーリング解析. *IBIS 2025*. 那覇文化芸術劇場なはーと.
9. 2025-09-22. (15:00-16:30) 司馬博文. メトロポリスを超えた枠組みで我々はどこまで行けるか？: PDMP と従来の MCMC はどう違い，どう違わないか？. **機械学習若手の会** (*YAML*) *2025*. ハートピア熱海.
10. 2025-08-28. (17:20-19:00) 司馬博文. サンプリングにおける加速と安定性. **連続最適化および関連分野に関する夏季学校**.
11. 2025-05-23. (13:40-14:20) 司馬博文. ベイズ変数選択の計算的解決: PDMP による非絶対連続分布からのサンプリング. **統計数理研究所オープンハウス**.
12. 2025-03-08. (12:00-12:50) 司馬博文. PDMP によりスパイク付きの非絶対連続分布からもサンプリングが可能. **第 *19* 回日本統計学会春季集会**. (優秀発表賞受賞)
13. 2025-02-17. (15:30-16:30) 司馬博文. ベイズ変数選択の計算的解決: PDMP による非可逆ジャンプの達成. **計算技術による学際的統計解析ワークショップ**.
14. 2024-05-24. (10:30-11:10) 司馬博文. 新時代の MCMC を迎えるために: 連続時間アルゴリズムへの進化. **統数研オープンハウス**.
15. 2024-02-25. (18:00-19:30) **Hirofumi Shiba**. A Recent Development of Particle Methods: Inquiry towards a Continuous Time Limit and Scalability. *MLSS2024 (OIST, Okinawa, Japan)*.

Research Visit

- School of Mathematics, **University of Bristol**. 2026.9.21 – 2026.10.21
  Supervisor: Prof. Christophe Andrieu
- Institute of Stochastics, **Ulm University**, Germany. 2025.7.13 – 2025.7.26
  Supervisor: Prof. Evgeny Spodarev
- Department of Statistics & Data Science, **National University of Singapore**. 2025.6.1 – 2025.6.30
  Supervisor: Dr. Alexandre Thiéry
- Department of Statistical Science, **University College London**, United Kingdom. 2024.11.4 – 2024.12.2
  Supervisor: Prof. Alexandros Beskos

Grants & Awards

- **Research Fellow**. *Graduate University for Advanced Studies, SOKENDAI, Tokyo, Japan*. 2025.4 – 2028.3

  Supported by JST BOOST, Japan Grant Number JPMJBS2412: *A Radical Approach to Trustworthy AI in the Coming Era*. ¥ 11,700,000.

- **Best Poster Award**, 19th Spring Meeting of the Japan Statistical Society. 2025.3.8
- **Director's Award** from the Japan Statistical Society Certificate. 2025.3.8

Service

Reviewing: Bayesian Analysis, Journal of Machine Learning Research

Additional Experience

- **Data Scientist**. *PreMedica, Inc., Tokyo, Japan*. 2024.9 – today

  Responsibilities: Provided Bayesian data analysis solutions to healthcare applications.

- **Research Assistant**. *The Institute of Statistical Mathematics, Tokyo, Japan*. 2023.7 – 2026.3

  Responsibilities: Contributed to the R package `YUIMA`, an open-source project aiming to simulate and infer multidimensional stochastic differential equations, with an emphasis on Bayesian inference.